

[K S Tharunpranav 251801359-2025-AIDS](#) ▾**K2****Started on** Thursday, 17 September 2026, 8:15 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:20 AM**Time taken** 4 mins 47 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  int upper_bound(int *arr, int low, int high, int val) {
4      while (low < high) {
5          int mid = low + (high - low) / 2;
6          if (arr[mid] <= val) {
7              low = mid + 1;
8          } else {
9              high = mid;
10         }
11     }
12     return low;
13 }
14 int main() {
15     int n;
16     if (scanf("%d", &n) != 1) return 0;
17     int *seq = (int *)malloc(n * sizeof(int));
18     for (int i = 0; i < n; i++) {
19         if (scanf("%d", &seq[i]) != 1) return 0;
20     }
21     int *dp = (int *)malloc(n * sizeof(int));
22     int len = 0;
23     for (int i = 0; i < n; i++) {
24         int idx = upper_bound(dp, 0, len, seq[i]);
25         dp[idx] = seq[i];
26         if (idx == len) {
27             len++;
28         }
29     }
30     printf("%d\n", len);
31     free(seq);
32     free(dp);
33     return 0;
34 }
35

```

|   | Input                 | Expected | Got |   |
|---|-----------------------|----------|-----|---|
| ✓ | 9<br>-1 3 4 5 2 2 2 3 | 6        | 6   | ✓ |
| ✓ | 7<br>1 2 2 4 5 7 6    | 6        | 6   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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